

Digital Computer Fundamentals

Complete student lesson for course code **3134**.

Revision 2026 Semester 3 Program Core 4 modules

Course snapshot

Scheme: 3 (L:3,T:0,P:0); 45 periods

Credits: 3

Contact: 3 (L:3,T:0,P:0)

Module hours listed: 43

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

3134

Semester

3

Credits

3

Teaching scheme

3 (L:3,T:0,P:0); 45 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Number systems and digital codes	8	CO1 · Applying
2	Boolean algebra and Karnaugh maps	11	CO2 · Applying
3	Combinational circuits	12	CO3 · Applying
4	Sequential circuits	12	CO4 · Applying

Prerequisites

- Fundamentals of Electrical and Electronics; semiconductor physics, diodes, transistors and rectifiers.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. Understand the data representation in the Computer System.
2. Perform number conversion from one system to another.
3. Understand use of boolean algebra and K-Map in digital circuits.
4. Equip students to design and develop simple combinational and sequential circuits.

Course Outcomes

1. Perform number system conversions, binary arithmetic operations and binary coding.
2. Make use of Boolean algebra and the Karnaugh Map for the implementation of logic functions.
3. Design combinational circuits.
4. Design sequential circuits.

CO-PO mapping as supplied

CO-PO mapping is reproduced from the supplied syllabus header; 3 = strongly mapped, 2 = moderately mapped, 1 = weakly mapped, and a blank cell is not silently converted into a value.

Complete Syllabus Coverage

Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

Source-to-lesson coverage map

Module	Lesson topic cards	Source basis	Official point units
Module 1	3	Official topic-row context	5
Module 2	3	Official topic-row context	5
Module 3	4	Official topic-row context	4
Module 4	4	Official topic-row context	4

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Number systems and base conversion	3
module-1	M1.02	Complements and binary arithmetic	3
module-1	M1.03	Binary codes and data representation	2
module-2	M2.01	Boolean laws and logic gates	4
module-2	M2.02	Canonical forms, minterms and maxterms	3
module-2	M2.03	Karnaugh-map simplification	4
module-3	M3.01	Half and full adders	3
module-3	M3.02	Subtractors and arithmetic circuits	3
module-3	M3.03	Encoders, decoders and comparators	3

Module	Topic code	Topic	Hours
module-3	M3.04	Multiplexers and demultiplexers	3
module-4	M4.01	Latches and flip-flops	3
module-4	M4.02	Registers and shift registers	3
module-4	M4.03	Counters and frequency division	3
module-4	M4.04	Timing and sequential-system design	3

Learning Roadmap

1. Number systems and digital codes

CO1 · Applying · 8 hours

2. Boolean algebra and Karnaugh maps

CO2 · Applying · 11 hours

3. Combinational circuits

CO3 · Applying · 12 hours

4. Sequential circuits

CO4 · Applying · 12 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Number systems and digital codes

Data representation is the foundation for digital hardware. The official content covers decimal, binary, octal and hexadecimal systems; conversions; signed and unsigned representations; complements; binary arithmetic; BCD, Gray and ASCII codes.

Hours: 8

CO1 · Applying · Bloom level: Applying

Official source coverage for Module 1

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M1.01 2 Applying system to different number systems Apply arithmetic operations on binary M1.02 2 Applying

M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding

M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding

M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying

M1.05 Solve addition of BCD numbers 1 Applying Contents:

Point-by-point study guide

– **Official syllabus point 1: M1.01 2 Applying system to different number systems Apply arithmetic operations on binary M1.02 2 Applying**

Exact source point

Syllabus text: M1.01 2 Applying system to different number systems Apply arithmetic operations on binary M1.02 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M1.01 2 Applying system to different number systems Apply arithmetic operations on binary M1.02 2 Applying”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.01 2 Applying system to different number systems Apply arithmetic operations on binary M1.02 2 Applying ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 2: M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding

Exact source point

Syllabus text: M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding

How to understand and study it

This point is an official syllabus requirement: “M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 3: M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding**

Exact source point

Syllabus text: M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding

How to understand and study it

This point is an official syllabus requirement: “M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 4: M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying**

Exact source point

Syllabus text: M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying

How to understand and study it

This point is an official syllabus requirement: “M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 5: M1.05 Solve addition of BCD numbers 1 Applying Contents

Exact source point

Syllabus text: M1.05 Solve addition of BCD numbers 1 Applying Contents

How to understand and study it

This point is an official syllabus requirement: “M1.05 Solve addition of BCD numbers 1 Applying Contents”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.05 Solve addition of BCD numbers 1 Applying Contents ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

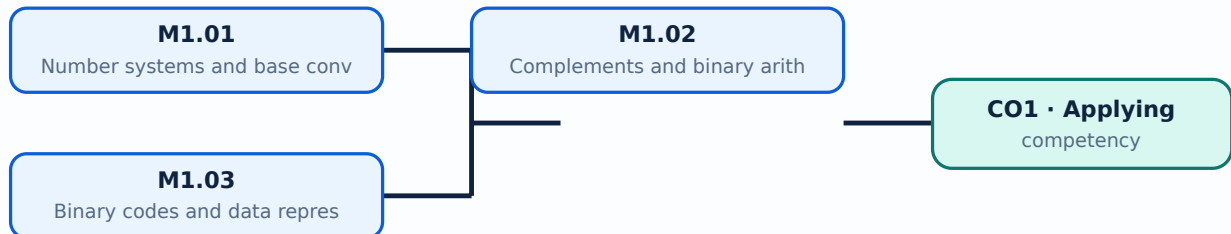
Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Learning goals

- Convert numbers between common bases.
- Perform binary arithmetic and identify suitable digital codes.
- Check a conversion by reversing the operation.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — Number systems and base conversion (3 hours)

Concept and explanation

A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.

Malayalam support: ഏകദേശം അളവുകൾക്ക് എങ്ങനെ മാറ്റം വരുത്തേണ്ടത് എന്നതിനെക്കുറിച്ച്. English technical terms അളവുകൾക്ക് മാറ്റം വരുത്തേണ്ടത്.

Study points

- Use positional weights.
- Convert integer and fractional values carefully.
- Write the radix when ambiguity is possible.

Engineering / workplace application: Microprocessors and memory addresses commonly use hexadecimal notation while logic circuits use binary states.

– M1.02 — Complements and binary arithmetic (3 hours)

Concept and explanation

One's complement and two's complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.

Malayalam support: ഏകപരിപൂർണ്ണതയും രണ്ടുപരിപൂർണ്ണതയും English technical terms complement, subtraction.

Study points

- Distinguish one's and two's complement.
- Check overflow in fixed-width arithmetic.
- Keep the word length consistent.

Illustrative example: For an n-bit two's-complement number, negate it by complementing all bits and adding 1, then discard a carry beyond the word length.

– M1.03 — Binary codes and data representation (2 hours)

Concept and explanation

BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.

Malayalam support: ബൈനറി കോഡ് English technical terms binary code.

Study points

- Explain BCD, Gray and ASCII.
- Convert a small value into a code.
- State one engineering use for each code.

Engineering / workplace application: Encoders, displays, keyboards and digital communication interfaces use coded representations.

Module summary: A reliable conversion method prevents errors when a digital quantity is represented in another base.

OFFICIAL MODULE 2

Boolean algebra and Karnaugh maps

Boolean algebra models logic with variables that take 0 or 1. The syllabus includes Boolean laws, logic gates, canonical forms, minterms, maxterms and Karnaugh-map simplification.

Hours: 11

CO2 · Applying · Bloom level: Applying

Official source coverage for Module 2

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3

M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding

M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1

M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying M2.05 2 Applying logic functions Series Test - I 1

Point-by-point study guide

– **Official syllabus point 1: M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3**

Exact source point

Syllabus text: M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3

How to understand and study it

This point is an official syllabus requirement: “M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 2: M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding

Exact source point

Syllabus text: M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding

How to understand and study it

This point is an official syllabus requirement: “M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.02 3 logic functions M2.03 Illustrate properties of basic, universal gates 3 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ് ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 3: M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1**

Exact source point

Syllabus text: M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1

How to understand and study it

This point is an official syllabus requirement: “M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.03 Illustrate properties of basic, universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 4: M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying**

Exact source point

Syllabus text: M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying ് technical terms English- ്. ് definition ് principle ്; ് term-ൿ function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Exact source point

Syllabus text: M2.05 2 Applying logic functions Series Test - I 1

How to understand and study it

This point is an official syllabus requirement: “M2.05 2 Applying logic functions Series Test - I 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.05 2 Applying logic functions Series Test - I 1 ് technical terms English-ൿ ്. ് definition ് principle ്; ് term-ൿ function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

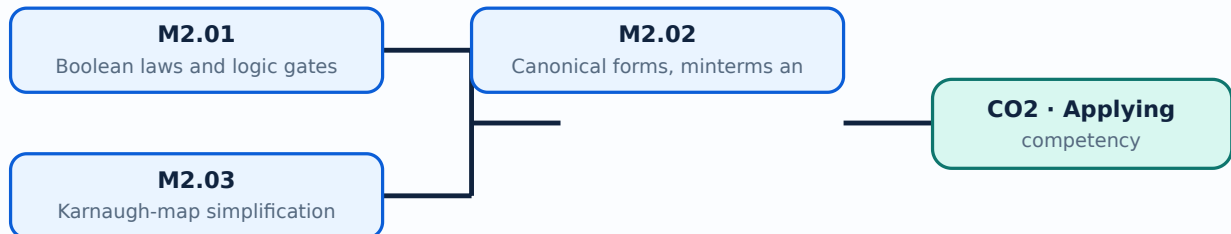
Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Learning goals

- Apply Boolean identities.
- Reduce a logic expression using a K-map.
- Translate between truth tables and expressions.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — Boolean laws and logic gates (4 hours)

Concept and explanation

Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan's theorems are especially useful for changing a circuit between active-high and active-low forms.

Malayalam support: ഓരോ തീർപ്പും ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ ഉപയോഗിച്ച് വിശദീകരിക്കുക. English technical terms use Malayalam words to explain.

Study points

- Use identity, null, idempotent and complement laws.
- Relate NAND and NOR to universal-gate implementation.
- Draw a truth table.

Engineering / workplace application: Combinational logic in controllers and interlocks is specified using Boolean expressions.

– M2.02 — Canonical forms, minterms and maxterms (3 hours)

Concept and explanation

A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Identify minterms from rows containing 1.
- Identify maxterms from rows containing 0.
- Use don't-care conditions only when justified.

Common mistake: Do not interchange the row conditions for minterms and maxterms.

– M2.03 — Karnaugh-map simplification (4 hours)

Concept and explanation

A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Make valid groups.
- Select essential prime implicants.
- Implement the reduced expression with suitable gates.

Illustrative example: For four variables, begin with the largest valid group, then cover every required 1-cell with the fewest essential groups.

Module summary: Simplification reduces gate count, delay and wiring in a practical digital circuit.

OFFICIAL MODULE 3

Combinational circuits

Combinational outputs depend only on present inputs. The official coverage includes adders, subtractors, comparators, encoders, decoders, multiplexers and demultiplexers.

Hours: 12

CO3 · Applying · Bloom level: Applying

Official source coverage for Module 3

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying
M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder to implement M3.03 2 Applying
M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for
M3.04 multiplication, comparison, multiplexing, 6 Applying decoding, encoding and priority encoding

Point-by-point study guide

– **Official syllabus point 1: M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying**

Exact source point

Syllabus text: M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.01 2 Understanding circuits M3.02 Construct half adder, full adder circuits 2 Applying ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 2: M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder to implement M3.03 2 Applying**

Exact source point

Syllabus text: M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder to implement M3.03 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder to implement M3.03 2 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.02 Construct half adder, full adder circuits 2 Applying Make use of binary parallel adder to implement M3.03 2 Applying ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 3: M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for

Exact source point

Syllabus text: M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for

How to understand and study it

This point is an official syllabus requirement: “M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 4: M3.04 multiplication, comparison, multiplexing, 6 Applying decoding, encoding and priority encoding

Exact source point

Syllabus text: M3.04 multiplication, comparison, multiplexing, 6 Applying decoding, encoding and priority encoding

How to understand and study it

This point is an official syllabus requirement: “M3.04 multiplication, comparison, multiplexing, 6 Applying decoding, encoding and priority encoding”. Prepare a comparison using the same criteria for every item: principle, performance, cost or complexity, limitation, and application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.04 multiplication, comparison, multiplexing, 6 Applying decoding, encoding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ് ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

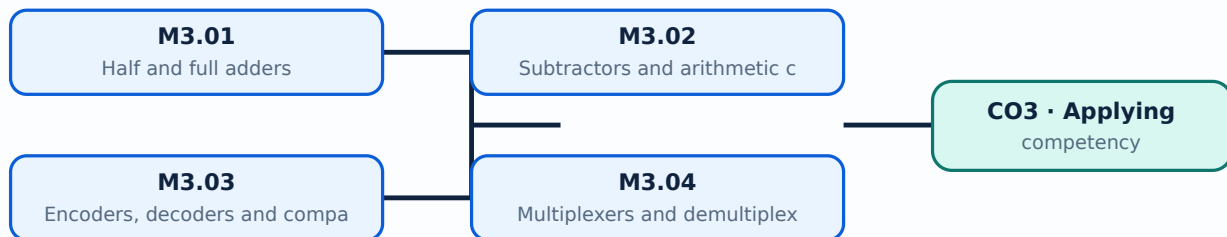
Learning goals

- Construct a truth table and logic expression.
- Choose a standard combinational block.

- Verify the output for representative input combinations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — Half and full adders (3 hours)

Concept and explanation

A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.

Malayalam support: ഏകദേശം 3 മണിക്കൂർ എങ്കിലും English technical terms ഉപയോഗിക്കുക.

Study points

- Write sum and carry equations.
- Explain carry propagation.
- Distinguish half and full adders.

Engineering / workplace application: Arithmetic logic units use cascaded full adders.

– M3.02 — Subtractors and arithmetic circuits (3 hours)

Concept and explanation

Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.

Malayalam support: ഏകപദം കണക്കാക്കുന്നതിനുള്ള സാങ്കേതിക വിദ്യകൾ English technical terms കണക്കാക്കുന്നതിനുള്ള സാങ്കേതിക വിദ്യകൾ.

Study points

- State difference and borrow outputs.
- Explain two's-complement subtraction.
- Check sign and overflow conditions.

– M3.03 — Encoders, decoders and comparators (3 hours)

Concept and explanation

An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.

Malayalam support: ഏകപദം കണക്കാക്കുന്നതിനുള്ള സാങ്കേതിക വിദ്യകൾ English technical terms കണക്കാക്കുന്നതിനുള്ള സാങ്കേതിക വിദ്യകൾ.

Study points

- Identify input/output roles.
- Use enable inputs correctly.
- Interpret priority encoders where more than one input may be active.

Engineering / workplace application: Address decoding and keypad interfaces use decoders and encoders.

– M3.04 — Multiplexers and demultiplexers (3 hours)

Concept and explanation

A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾക്ക് ഇംഗ്ലീഷ് പദങ്ങൾ ഉപയോഗിക്കുക. English technical terms use Malayalam words.

Study points

- Calculate the number of select lines.
- Trace the selected data path.
- Use a multiplexer to realise a Boolean function.

Module summary: A clean design proceeds from specification to truth table, simplification, circuit drawing and verification.

OFFICIAL MODULE 4

Sequential circuits

Sequential circuits include memory, so output depends on present inputs and previous state. The syllabus covers latches, flip-flops, registers and counters.

Hours: 12

CO4 · Applying · Bloom level: Applying

Official source coverage for Module 4

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops

M4.02 4 Applying flip flops M4.03 Outline different type of registers 1

Understanding

M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying

M4.04 6 Applying counters using flip flops Series Test - II 1

Point-by-point study guide

Understanding Illustrate operation of different latches and flip flops

Syllabus text: M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□ M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches, M4.02 4 Applying flip flops □□□□ technical terms English-□□□□□□ □□□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 2: M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding

Exact source point

Syllabus text: M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding

How to understand and study it

This point is an official syllabus requirement: “M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 3: M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying**

Exact source point

Syllabus text: M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying

How to understand and study it

This point is an official syllabus requirement: “M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.03 Outline different type of registers 1 Understanding Construct asynchronous, synchronous M4.04 6 Applying ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– Official syllabus point 4: M4.04 6 Applying counters using flip flops Series Test - II 1

Exact source point

Syllabus text: M4.04 6 Applying counters using flip flops Series Test - II 1

How to understand and study it

This point is an official syllabus requirement: “M4.04 6 Applying counters using flip flops Series Test - II 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.04 6 Applying counters using flip flops Series Test - II 1 ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

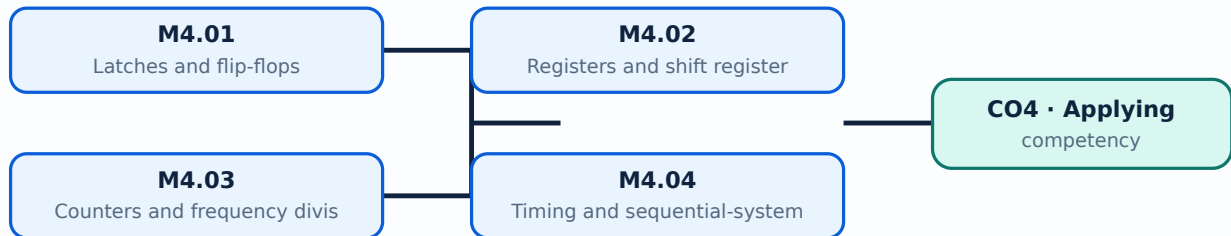
Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Learning goals

- Explain the role of a clock and state.
- Read characteristic and excitation tables.
- Select a register or counter arrangement.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.01 — Latches and flip-flops (3 hours)

Concept and explanation

A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.

Malayalam support: ഞങ്ങൾ ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾക്ക് മലയാളത്തിൽ പരിഭാഷ നൽകുന്നു. English technical terms മലയാളത്തിൽ പരിഭാഷ നൽകുന്നു.

Study points

- Compare level and edge triggering.
- Use preset and clear safely.
- Read a characteristic table.

– M4.02 — Registers and shift registers (3 hours)

Concept and explanation

A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.

Malayalam support: റിജിസ്റ്റർ ഷിഫ്റ്റ് റിജിസ്റ്റർ എന്നിവയാണ്. English technical terms റിജിസ്റ്റർ ഷിഫ്റ്റ് റിജിസ്റ്റർ.

Study points

- Identify data direction.
- Explain serial and parallel transfer.
- Relate a register to temporary storage.

Engineering / workplace application: Serial communication and delay lines use shift registers.

– M4.03 — Counters and frequency division (3 hours)

Concept and explanation

Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.

Malayalam support: കൗണ്ടർ ഫ്രീക്വൻസി ഡിവിഷൻ എന്നിവയാണ്. English technical terms കൗണ്ടർ ഫ്രീക്വൻസി ഡിവിഷൻ.

Study points

- Distinguish asynchronous and synchronous counters.
- Determine modulus from the state sequence.
- Explain reset and unused states.

– M4.04 — Timing and sequential-system design (3 hours)

Concept and explanation

A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Malayalam support: ഞങ്ങൾ ഇംഗ്ലീഷ് ടെക്നിക്കൽ ടേർമിനോളജി ഉപയോഗിക്കുന്നു. English technical terms are used.

Study points

- Draw a state diagram.
- Check timing constraints.
- Use a reset state for predictable startup.

Module summary: Sequential design requires attention to timing, state transitions, setup, hold and reset behaviour.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

Open the module to view its labelled learning-map diagram.	Module 2: Boolean algebra
Open the module to view its labelled learning-map diagram.	Module 3: Combinational
Open the module to view its labelled learning-map diagram.	Module 4: Sequential
Open the module to view its labelled learning-map diagram.	

Official Activities and Practical Requirements

Official activities / self-learning

- No separate activity list is provided in the supplied PDF.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Series Test: 2 periods/assessment component as stated in the supplied syllabus; the supplied header does not state a separate CIE/ESE mark distribution.

Module-wise Questions and Answers

module-1 — Number systems and digital codes

1. Explain Number systems and base conversion. [3 marks]

A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Complements and binary arithmetic. [3 marks]

One's complement and two's complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Binary codes and data representation. [3 marks]

BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Boolean algebra and Karnaugh maps

4. Explain Boolean laws and logic gates. [3 marks]

Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan's theorems are especially useful for changing a circuit between active-high and active-low forms.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

5. Explain Canonical forms, minterms and maxterms. [3 marks]

A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Karnaugh-map simplification. [3 marks]

A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Combinational circuits

7. Explain Half and full adders. [3 marks]

A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Subtractors and arithmetic circuits. [3 marks]

Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

9. Explain Encoders, decoders and comparators. [3 marks]

An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Multiplexers and demultiplexers. [3 marks]

A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Sequential circuits

11. Explain Latches and flip-flops. [3 marks]

A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Registers and shift registers. [3 marks]

A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

13. Explain Counters and frequency division. [3 marks]

Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Timing and sequential-system design. [3 marks]

A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. Define or explain *Number systems and base conversion* with one relevant application. (5 marks)
2. Define or explain *Complements and binary arithmetic* with one relevant application. (5 marks)
3. Define or explain *Boolean laws and logic gates* with one relevant application. (5 marks)
4. Define or explain *Canonical forms, minterms and maxterms* with one relevant application. (5 marks)
5. Define or explain *Half and full adders* with one relevant application. (5 marks)
6. Define or explain *Subtractors and arithmetic circuits* with one relevant application. (5 marks)
7. Define or explain *Latches and flip-flops* with one relevant application. (5 marks)
8. Define or explain *Registers and shift registers* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Number systems and digital codes:** A reliable conversion method prevents errors when a digital quantity is represented in another base.
- **Boolean algebra and Karnaugh maps:** Simplification reduces gate count, delay and wiring in a practical digital circuit.

- **Combinational circuits:** A clean design proceeds from specification to truth table, simplification, circuit drawing and verification.
- **Sequential circuits:** Sequential design requires attention to timing, state transitions, setup, hold and reset behaviour.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Number systems and base conversion	A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.
Complements and binary arithmetic	One's complement and two's complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.
Binary codes and data representation	BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.
Boolean laws and logic gates	Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan's theorems are especially useful for changing a circuit between active-high and active-low forms.
Canonical forms, minterms and maxterms	A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

Term	Short meaning
Karnaugh-map simplification	A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.
Half and full adders	A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.
Subtractors and arithmetic circuits	Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.
Encoders, decoders and comparators	An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.
Multiplexers and demultiplexers	A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.
Latches and flip-flops	A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.
Registers and shift registers	A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.
Counters and frequency division	Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.
Timing and sequential-system design	A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Official References and Resources

References listed in the supplied syllabus

1. Digital Design, M. Morris Mano.
2. Fundamentals of Digital Circuits, A. Anand Kumar.

Official learning resources listed

- <https://www.w3schools.com/>

In-page Search

Generated as a student lesson from the supplied syllabus PDF. Revision: Revision 2026 · Course code: 3134. Practice questions and explanations are educational support, not official examination material.